Reproducing and Extending A Novel Hybrid Approach For Efficiently Forecasting Air Quality Data

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Abstract

Air quality changes over several time scales, from short pollution spikes to slower daily and seasonal patterns. This project studies WaveCatBoost, which separates a pollutant series into wavelet components, predicts each component with CatBoost, and combines the predictions. We implement the main idea in R and examine saved experiments on the paper's CPCB and ID1 data. We then extend the work with alternative boosting models and wavelet filters, a cleaned CPCB comparison, and additional Shillong station studies. All numerical results in this report come from project experiments rather than tables copied from the assigned paper. Across 48 original-data cases, WaveCatBoost has a mean MASE of 1.103, compared with 1.111 for plain CatBoost, but the ranking depends on the summary measure and forecasting setting. In the cleaned, direct 24-hour CPCB experiment, separate component models achieve a test MAE of 7.480, while a simple hour-of-day baseline achieves 7.059; paired comparisons do not establish a clear winner among the five learned architectures. The retained Shillong study suggests that other pollutants' histories can be more useful than decomposition alone, although its training code is incomplete and some design choices used test-period information. The main lesson is that data quality, time-aware evaluation, and simple baselines matter as much as model complexity. Wavelet forecasting is a useful option, but the available evidence does not support treating it as the best method for every station or horizon.

1. Introduction

Air-quality forecasting asks a practical question: given the measurements available now, what concentration should we expect later? Concentrations contain short spikes, daily patterns, and slower seasonal changes. Missing readings and sensor faults make the task harder: a regular-looking series is not necessarily a reliable record.

Our assigned paper, Borah et al. (2025), proposes WaveCatBoost. It separates a pollutant series into faster and slower wavelet components, fits CatBoost to each part, and combines the forecasts. A conformal prediction layer adds uncertainty bands. We reproduce this modelling idea in R and investigate whether its extra complexity is useful on the available data.

We ask three questions: does decomposition improve ordinary boosting; does the answer change when invalid readings are excluded; and can shared models or other pollutants provide useful extensions? We compare the saved CPCB and ID1 experiments, a cleaned CPCB study, and additional Shillong results. The report also records evaluation limits, because low error can reflect artificial fills, easier test windows, or future information in preprocessing.

This is a reproduction of the method rather than an exact numerical replication of the published tables. Our original-data benchmark uses recursive forecasts from one terminal origin. The companion workflow also includes rolling one-step evaluation. These answer different questions, so we report project measurements and keep their protocols separate.

2. Background and Related Work

2.1. Wavelets and component forecasting

Wavelets describe changes at different time scales. Fine components record rapid variation; coarse components capture slower movement. MODWT retains aligned coefficients at every timestamp, which makes lag features convenient (Percival & Walden, 2000). Its components are not automatically independent, and rapid changes are not necessarily noise. Component forecasting also adds fitting cost and makes boundary handling important: a transform of a completed series may use information unavailable at an earlier forecast origin.

2.2. Boosted trees and simple references

CatBoost combines small decision trees and uses ordered boosting to reduce a particular training prediction shift (Prokhorenkova et al., 2018). It does not replace chrono-

logical validation or causal features. XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) provide alternative learners for the same lag-based setup. Seasonal naive, SARIMA, persistence, and historical averages test whether the added complexity is worthwhile.

2.3. Accuracy and uncertainty

MAE measures a typical absolute error; RMSE emphasises large misses. MASE divides MAE by a naive error estimated on training data (Hyndman & Koehler, 2006). It supports comparisons across scales, but becomes sensitive when training variation is small.

Conformal prediction uses past forecast errors to build prediction intervals. Standard finite-sample coverage arguments require assumptions such as exchangeability of calibration and future scores (Angelopoulos & Bates, 2023). Air-quality time series are dependent and can change with the season. A nominal 95% interval therefore needs a separate coverage check; the label alone is not evidence that it works.

3. Approach

3.1. Problem definition and input windows

Let y_t be an hourly concentration and X_t contain information available by t. A direct model predicts y_{t+h} using a model specific to horizon h. A recursive model predicts one hour, appends the prediction to its history, and repeats. The original-data comparison is recursive; cleaned CPCB uses direct prediction at $h = 24$ hours. Its 24-value input window runs from the origin t back to $t - 23$. No future target is an input.

3.2. Wavelet decomposition and reconstruction

For an additive multiresolution representation,

$$y_t = \sum_{j=1}^{J} D_{j,t} + S_{J,t}. \tag{1}$$

The detail series D_j describe changes at different scales, and S_J is the remaining smooth part. We cap J at five to control fitting cost, producing six component models. This is an implementation choice rather than a theoretical optimum.

Each component is converted into a lag table for regression. Recursive prediction uses earlier forecasts after observed history runs out. Forecasts of additive MRA components are summed to recover the level; raw MODWT coefficients would require their synthesis step instead.

The legacy periodic transform excludes the test values, but reconstructed training components can depend on later training observations and wrap around the boundary. Test exclusion alone therefore does not make every historical training feature causal.

3.3. CatBoost fitting and alternatives

CatBoost maps 24 component lags to the next value. The selected original-data setting requests ordered boosting, depth six, preceding-tail validation, and a maximum of 120 rounds. LightGBM and XGBoost use analogous lag tables with their own learners and objectives. The nine-method comparison includes these three plain models, seasonal naive and SARIMA, plus four wavelet variants. We did not train new RNN, TCN, TFT, or Transformer baselines; their published numbers are not mixed with our results.

3.4. Prediction intervals

The paper's conformal interval builds a band around the WaveCatBoost point forecast. Absolute forecast error is first scaled by $\widehat{U}_t$; a trailing-window quantile of those scores then sets the half-width at a new origin:

$$CS_t = \frac{|Y_t - \widehat{Y}_t|}{\widehat{U}_t}, \tag{2}$$

$$CQ_t = \inf\Big\{q : \tfrac{1}{\min(\kappa, t-1)} \textstyle\sum_{t'=1}^{t-1} CS_{t'}\, w_{t'} \geq 1 - \alpha\Big\}, \tag{3}$$

$$w_{t'} = \mathbf{1}(t' \geq t - \kappa), \tag{4}$$

$$\widehat{U}_t = \tfrac{1}{u} \textstyle\sum_{j=0}^{u-1} |e_{t-j}|, \quad e_s = Y_s - \widehat{Y}_s, \quad u = 24, \tag{5}$$

$$PI_t = \big[\widehat{Y}_t \pm CQ_t\, \widehat{U}_t\big]. \tag{6}$$

Here CS_t is the conformal score (nonconformity), CQ_t is the $(1-\alpha)$-quantile of the last κ scores, $w_{t'}$ restricts that average to the trailing window, $\widehat{U}_t$ is the rolling mean absolute residual, and PI_t is the resulting interval. We use $\alpha = 0.05$ (95% nominal coverage) and choose $\kappa \in \{24, 168, 336\}$ hours on calibration. The quantile CQ_t must be estimated before seeing the test targets.

4. Extensions Beyond the Assigned Paper

4.1. Alternative learners and filters

We replace CatBoost with LightGBM and XGBoost and compare Haar, D4, and LA8 filters. These experiments separate the effect of decomposition from the learner and filter. Their scores come from separate runs, not the main comparison cache. Fourier-feature and empirical mode decomposition trials have narrower retained coverage and are excluded from the headline ranking.

4.2. Cleaning before forecasting

The cleaned CPCB store has 8,880 hours after removing the final cloned CSV row. Five repeated six-pollutant vectors flag 2,637 likely synthetic hours (29.7%), including 821 test hours. These are pattern-based quality flags, not laboratory confirmation of individual faults.

Training-learned repeated-vector rules, physical bounds, stuck runs, and the $PM_{2.5}$–PM_{10} consistency rule mark invalid readings as missing. Values are not clipped. Past-only fills supply model inputs, while invalid or filled targets are excluded from scoring. A trailing 2,048-hour decomposition retains only each window's endpoint, so historical components use information available at their own timestamps.

4.3. Separate, shared, and multivariate models

B0 fits six separate own-component models. A1 predicts the raw level from target lags and other pollutants at the origin. B1 shares one learner across components; B2 adds other-pollutant components to separate learners; B3 combines sharing and those added inputs. Shared models use a component identifier and training standardisation, with standard-deviation weights restoring original-unit absolute-error weighting. Validation chooses same-scale or all-scale covariates for B2/B3.

The unmasked control trains on the original series but uses the same retained target timestamps and observed truth. It isolates cleaning from a change in test cases.

4.4. Additional Shillong studies

The retained Lumpyngngad and JN Stadium study compares raw, multivariate, decomposed, shared, and covariate-augmented models at 24, 168, 336, and 744 hours. Its architecture names are study-specific: Shillong A1 means raw target lags plus covariates. Reports and aggregate CSVs remain, but the training runner, configuration, and some second-stage code are missing. These imported results cannot be refitted end to end. The separate short-horizon Shillong runner follows a different protocol and is not pooled with them.

5. Assumptions and Analytical Observations

We do not claim a new forecasting theorem. Two simple observations explain the design and its limits. First, suppose Equation 1 holds at the target hour. If e_j is the error of the forecast for component j, then the reconstructed forecast error is the sum of those component errors. Consequently,

$$|e_{\text{level}}| \leq \sum_{j=1}^{J+1} |e_j|. \tag{7}$$

This follows directly from the triangle inequality. It gives an upper bound, not a promise that decomposition improves accuracy. Errors can reinforce or cancel. Better detail-component scores alone therefore do not establish a better reconstructed forecast.

Second, causal preprocessing requires every feature at origin t to be a function of observations available by t. A backward-looking coefficient satisfies this condition if its fitted parameters also come only from permitted earlier data. A two-sided transform of a completed series generally does not. The same argument applies to imputation, standardisation, and pollutant selection. Ordered boosting cannot repair an input that already contains future information.

For conformal intervals, any claim of finite-sample nominal coverage also needs the calibration assumptions of the chosen procedure. The observed seasonal shifts and flat calibration periods in this project make those assumptions questionable. We therefore report empirical failures and avoid interpreting the intervals as guaranteed bounds for future air quality.

6. Experimental Setup

6.1. Data and evidence sources

The public CPCB and ID1 hourly CSVs contain 8,881 and 9,505 rows and six pollutants (Borah, 2025). Cleaned CPCB has 8,880 hours from 1 July 2022 to 5 July 2023. Units are retained as recorded; uncertain CO conventions prevent direct physical-unit comparisons across pollutants.

Project outputs supply every reported result (ATS Project Group 8, 2026). We restrict `results/paper/scores.csv` to CPCB and ID1: 432 rows across nine methods, two sensors, six pollutants, and four windows. The extra SHILLONG group and illustrative values are excluded.

Cleaned predictions and comparisons are in `results/preprocessed_experiment/`; imported Shillong evidence is in `results/shillong/architecture/`. We checked current files without retraining models.

6.2. Original-data forecasting protocol

Original-data runs recursively predict the final 24, 168, 336, or 744 hours from one origin each. These nested windows are dependent and combine horizon with averaging difficulty; a longer window can have lower average error.

Settings selected on CPCB's one-day split overlap the longer test suffixes. Together with periodic boundaries and synthetic fills, this makes the legacy ranking descriptive rather than a clean significance test.

6.3. Cleaned direct 24-hour protocol

Training targets end on 28 February 2023 at 23:00. Validation targets run from 1 March to 30 April, and test targets run from 1 May to 5 July. Splits are based on the target timestamp $t + 24$, preventing a training origin near the boundary from contributing a later-block target. After input and target eligibility checks, all five architectures use 2,496 training rows, 757 validation targets, and 658 test targets. Final refitting uses 3,367 eligible train-plus-validation rows.

LightGBM uses L1 loss, seed 7, deterministic settings, one thread, feature fraction 0.8, up to 3,000 rounds, and early stopping after 100 rounds without improvement. The grid contains 18 configurations: leaves in $\{15, 31, 63\}$, learning rate in $\{0.03, 0.1\}$, and minimum leaf size in $\{20, 50, 100\}$. Validation chooses settings and iteration counts. These choices are frozen before refitting and predicting the test block. Validation accuracy is a selection score, not an unbiased final estimate.

The seasonal MASE denominator uses valid training pairs separated by 24 hours and equals 4.356. Scalers are estimated on training data and updated on train plus validation for final fitting. B0 and B2 tune components by component MAE; B1 and B3 use reconstructed $PM_{2.5}$ MAE. Uncertainty in paired model differences is estimated with a moving-block bootstrap of block length 24 and 2,000 repetitions. These intervals describe this test sample under that resampling choice, not all possible stations or seasons.

6.4. Shillong protocol and limits

The retained architecture report describes 15-minute readings aggregated to hourly values using at least three valid slots out of four. Lumpyngngad uses January–July for training, August–September for validation, and October–December for testing. JN Stadium starts reporting in late July; it uses late July–September for training, October for validation, and November–December for testing. All dates refer to 2023, with an end-labelled final hour at 1 January 2024.

The study uses direct horizon-specific models and common eligible targets across its architectures. LightGBM has learning rate 0.05, leaves in $\{15, 31\}$, minimum leaf size in $\{20, 100\}$, and up to 2,000 rounds with 100-round early stopping. Reported first-stage forecasts average seeds 2026–2028. A second stage explores other learners on shortlisted designs, but we restrict Table 4 to first-stage LightGBM results with retained aggregate evidence.

There is an additional limitation beyond missing training code: pollutant availability was screened using all three blocks, and Lumpyngngad ozone was excluded after a test-period regime change was observed. The resulting comparison is exploratory and informed by test inspection. A new untouched period is needed before treating its model choices as validated deployment decisions.

Table 1. Measured original-data results, CPCB and ID1 only. MASE is summarised across 48 cases per method. Time is the median saved elapsed seconds per case; it is indicative rather than a controlled hardware benchmark.

Method	Mean	Median	Time (s)
WaveCatBoost	1.103	0.948	95.83
CatBoost	1.111	0.921	15.34
LightGBM	1.174	0.961	1.12
SARIMA	1.182	1.008	19.56
XGBoost	1.199	1.035	1.51
WaveSARIMA	1.223	1.026	5.08
WaveXGB	1.238	1.076	9.38
WaveLGB	1.240	0.970	7.64
Seasonal naive	1.262	1.105	0.02

7. Results and Discussion

7.1. Original-data model comparison

Table 1 summarises the 48 cases for each method. WaveCatBoost has the smallest mean MASE, 1.103, but plain CatBoost is close at 1.111 and has the better median, 0.921 versus 0.948. The mean difference is about 0.008 MASE, or 0.7% relative to plain CatBoost. These descriptive results do not establish a reliable overall improvement, especially given the overlapping windows and selection limits.

WaveCatBoost also costs more to fit: its median saved time is 95.83 seconds, compared with 15.35 seconds for CatBoost. WaveLGB is faster than WaveCatBoost at 7.65 seconds, but its mean MASE is higher at 1.240. Thus neither "wavelets always improve boosting" nor "LightGBM gives the same accuracy for less time" is supported by this comparison.

The $PM_{2.5}$ cases in Table 2 show why a single average is incomplete. At CPCB's one-day origin, WaveCatBoost scores 0.712 against CatBoost's 0.821. At 31 days, WaveLGB scores 0.572 against LightGBM's 2.279. However, most CPCB methods are close to 1.50 over the seven-day suffix. At ID1's 31-day origin, seasonal naive scores 0.548 and WaveCatBoost 0.551, giving almost no advantage to the complex model. Different windows expose different behaviours.

7.2. Filter choice

In the separate WaveCatBoost filter experiment, CPCB $PM_{2.5}$ one-day MASE is 0.727 for Haar, 0.909 for D4, and 0.955 for LA8. On ID1 $PM_{2.5}$, the corresponding values are 0.421, 0.457, and 0.502. Haar is therefore a reasonable simple choice in these runs. It is not the winner in

Table 2. Measured recursive $PM_{2.5}$ MASE on the original data. The day column gives the length of the terminal forecast path. LGB and Cat denote plain models; WaveLGB and WaveCat use decomposition. These overlapping windows are descriptive cases, not independent test replications.

Sensor	Days	Seasonal naive	LGB	Cat	WaveLGB	WaveCat
CPCB	1	2.021	0.737	0.821	1.031	0.712
CPCB	7	1.496	1.502	1.499	1.497	1.499
CPCB	14	0.734	0.749	0.775	0.735	0.754
CPCB	31	2.871	2.279	2.393	0.572	1.056
ID1	1	0.710	0.844	0.672	0.360	0.415
ID1	7	0.751	0.736	0.619	0.540	0.590
ID1	14	0.758	0.654	0.686	0.691	0.681
ID1	31	0.548	0.569	0.549	0.738	0.551

Table 3. Cleaned CPCB direct 24-hour test results. MASE uses the training scale 4.356. Baselines use the same observed targets as the learned models.

Model	MAE	RMSE	MASE
B0	7.480	10.108	1.717
A1	7.842	10.470	1.800
B1	7.591	10.292	1.743
B2	8.062	9.917	1.851
B3	7.726	9.939	1.774
Persistence	8.134	11.709	1.867
Training mean	7.382	11.250	1.695
Hour-of-day mean	7.059	10.694	1.621

every pollutant case: CPCB NO_2 is slightly better with D4, while CPCB SO_2 is better with LA8. Filter choice should be checked on a preceding validation period rather than declared universal.

7.3. Cleaned CPCB architecture comparison

Table 3 gives the final test results on the same 658 valid targets. B0 has the lowest MAE among the five learned architectures, at 7.480. B2 has the lowest RMSE among those architectures, at 9.917. These are different objectives: B2 can reduce large misses without giving the lowest average absolute error.

The hour-of-day baseline has a lower test MAE than every learned architecture, while B0 has lower RMSE than that baseline. This is an important practical result. A six-model system is not automatically preferable when a simple historical average is competitive. The intended loss and the importance of large errors should guide the decision.

Validation does not preserve the test ordering. B3 has validation MAE 6.972 and B0 has 7.257, but B0 is better on the test block. Figure 3 makes this reversal visible. Because validation was used for tuning, its best score is optimistic as an estimate of future performance.

Paired comparisons do not show a clear architecture winner. The test MAE difference A1 minus B0 is $+0.36$, with a 95% bootstrap interval of $[-0.37, 0.98]$. For B1 minus B0 it is $+0.11$, with interval $[-0.21, 0.30]$. B2 minus B0 is $+0.58$, with interval $[-0.19, 1.36]$. Each interval includes zero. The shared B1 model is close to the six-model B0 result, so sharing may be attractive for simplicity, but the evidence does not establish that it improves accuracy.

7.4. What cleaning changed

Cleaning reduced every architecture's validation MAE, but did not consistently improve the final test scores. B0 improves from 7.757 in the unmasked control to 7.480 after cleaning, a difference of -0.277 whose bootstrap interval includes zero. A1 becomes worse, from 7.416 to 7.842; the difference is about $+0.43$, with interval $[0.17, 0.66]$. A better-looking validation table was therefore not enough to predict a general test improvement.

This does not make artificial fill values suitable targets. Cleaning improves the meaning of the experiment by excluding suspect observations from ground truth. Accuracy and validity are different questions. A model can obtain a convenient error by predicting repeated fill patterns, even if it is less useful on actual sensor measurements.

All cleaned test MASE values exceed one because they use the training seasonal-naive error as their scale. That does not mean every model loses to persistence on this test block: B0's test MAE is 7.480, compared with persistence's 8.134. It means that test errors are larger than the naive error measured during training. Seasonal changes make that distinction essential.

7.5. Retained Shillong evidence

Table 4 shows selected LightGBM results from the imported study. At Lumpyngngad, using the histories of other retained pollutants (A2) improves on raw univariate lags (A0) at every reported horizon. The largest reduction is at 168 hours, from 15.64 to 10.94. Decomposition alone (B0) does not give a consistent gain and is worse at 744 hours,

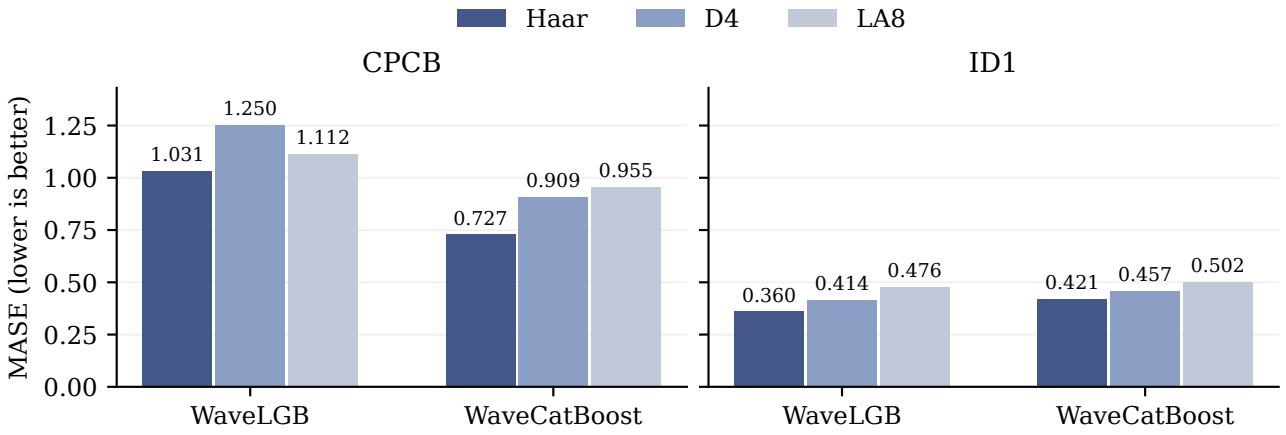


Figure 1. One-day $PM_{2.5}$ MASE for Haar, D4 and LA8. The same filter experiment compares WaveLGB and WaveCatBoost at each sensor; labels give the measured scores.

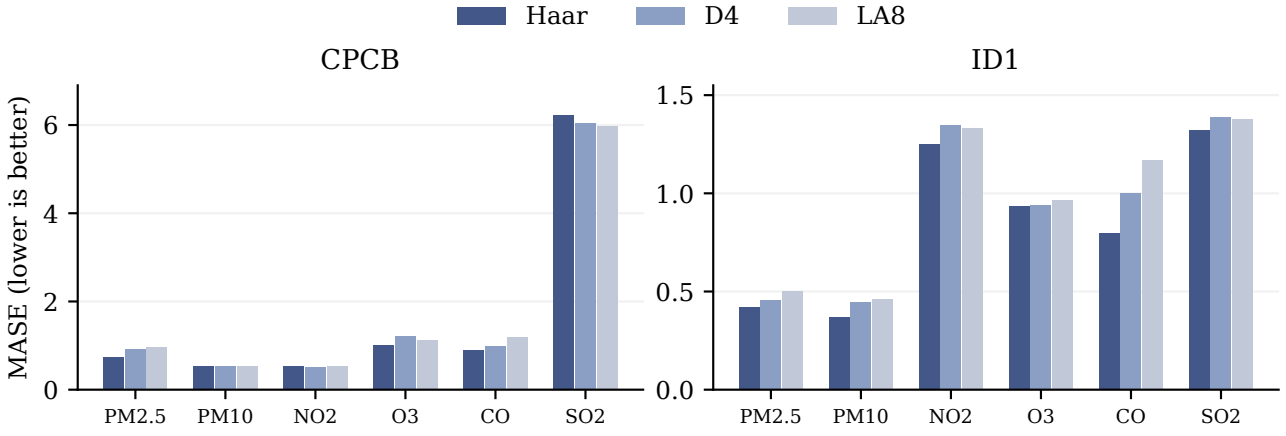


Figure 2. One-day WaveCatBoost MASE for all six pollutants and three filters. Each sensor has its own vertical scale. Haar is often competitive, but D4 and LA8 improve selected pollutant cases.

Table 4. Retained Lumpyngngad study: test $PM_{2.5}$ MAE. A0 is raw univariate, A2 uses multivariate lags, and B0 has separate univariate component models. Results are exploratory; see the missing-code and test-inspection limits in Section 6.4.

Model	24 h	168 h	336 h	744 h
A0	9.84	15.64	16.82	13.10
A2	9.11	10.94	12.29	11.42
B0	10.38	14.21	16.09	15.69
Persistence	10.03	13.28	15.70	14.65

15.69 versus A0's 13.10.

JN Stadium is less conclusive. At 24 hours, A0 has test MAE 23.77, A2 has 23.88, and persistence has 21.57. Its short record covers only a few months, and the test season differs from training. The two stations therefore do not support one universal architecture ranking. The study's ozone exclusion also followed test inspection, so even the promising Lumpyngngad multivariate results require confirmation on fresh data.

7.6. Uncertainty and evaluation lessons

The retained audit reports roughly 30.4% interval coverage for CPCB and 100% for ID1 in the separate seven-day calibration experiment, despite a nominal 95% target. CPCB's calibration tail is flat and produces a narrow band; ID1's band is much wider. These findings are evidence of calibration problems, not successful uncertainty quantification. Bands estimated from the test residuals are excluded from any coverage claim.

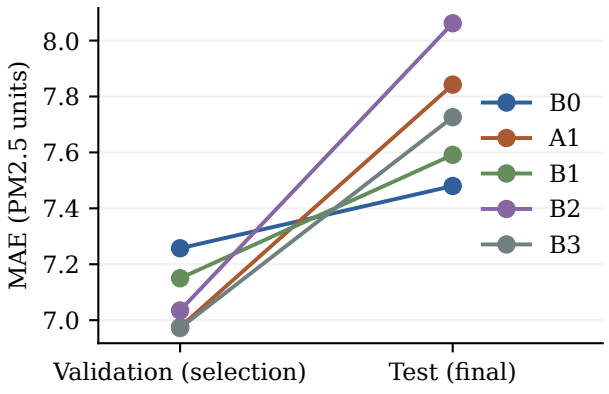


Figure 3. Cleaned CPCB MAE before and after the final temporal split. Lower is better. The architecture ranking changes between validation and test. Lines connect scores, not forecasts over time.

We also avoid the historical multiple-comparisons significance claim. Its saved critical-distance calculation is inconsistent with the current number of cases, and the nested horizon windows are dependent. A fresh comparison should use several forecast origins, preserve temporal dependence in its uncertainty estimates, and freeze preprocessing and model selection before the final test period.

7.7. Current recorded conformal experiment

The updated repository retains a later rolling one-step split-conformal cache in `results/paper/conformal.rds`. Table 5 and Figure 4 report this cache directly, with coverage and width recomputed from its saved bounds. This is a separate run from the older recursive-calibration audit described above. Each row covers 168 test hours after a 168-hour calibration block. Its time-varying scale uses errors observed before each successive test prediction; it is not a fixed seven-day-ahead interval. Point errors are specific to these one-step predictions and must not be pooled with the recursive scores in Table 2. Nominal 95% coverage is not attained on any of the three retained windows.

8. Reproducibility and Limitations

Run the experiments from the repository root. The original-data runner is `scripts/experiments/run_experiments.R`; filter comparisons use `scripts/experiments/run_filter_ablation_catboost.R`. For cleaned CPCB, run `scripts/preprocessing/run_preprocess.Rpaper`, followed by `scripts/experiments/run_preprocessed_experiment.R` with `phase1`, `phase2`, `phase3`, and `phase4`, in order. The repository README provides the complete commands and result paths.

The cleaned experiment settings are in `config/preprocessed_experiment.yaml` and `config/decomp.yaml`. Preserve the recorded caches before refitting, and save software versions, seeds, input hashes, and settings with new results. CatBoost must be installed for its scores; the implementation otherwise falls back to XGBoost. A cached run is not a fresh reproduction, and some legacy summaries lack compatible retained predictions.

The cleaned study covers one station and one seasonal test block. The imported Shillong study has incomplete training code and test-informed design choices. Together with uncertain units and sensor quality, these limits prevent a broad deployment claim.

9. Conclusion and Future Work

WaveCatBoost offers a clear modelling idea: separate time scales, forecast the parts, and combine them. Our original-data runs are competitive but show only a small mean advantage over plain CatBoost. In cleaned CPCB, no learned architecture is clearly best, and an hour-of-day average has the lowest test MAE. The Shillong evidence favours useful cross-pollutant information over a consistent benefit from decomposition alone.

The central lesson is that data and evaluation matter as much as model choice. Artificial fills change the target, boundary filters can use future information, and validation rankings can reverse in the next season. MASE must be read alongside its training scale and a test-period baseline.

Future work should use several rolling origins, causal preprocessing, several years per station, and an untouched final period. Raw and decomposed models should receive equivalent available information. Weather inputs must reflect what was known at the origin, and interval coverage should be checked separately by horizon and season.

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Table 5. Current recorded one-step split-conformal diagnostics for $PM_{2.5}$. Calibration and test coverage are percentages; width and MAE are in each series' source units. The target nominal coverage is 95%. Bounds are clipped at zero in the retained implementation.

Sensor	κ (h)	Cal. cover (%)	Test cover (%)	Mean width	Point MAE
CPCB	24	100.0	67.9	8.63	4.71
ID1	168	94.6	90.5	17.35	4.31
Shillong	168	94.6	90.5	127.25	50.46

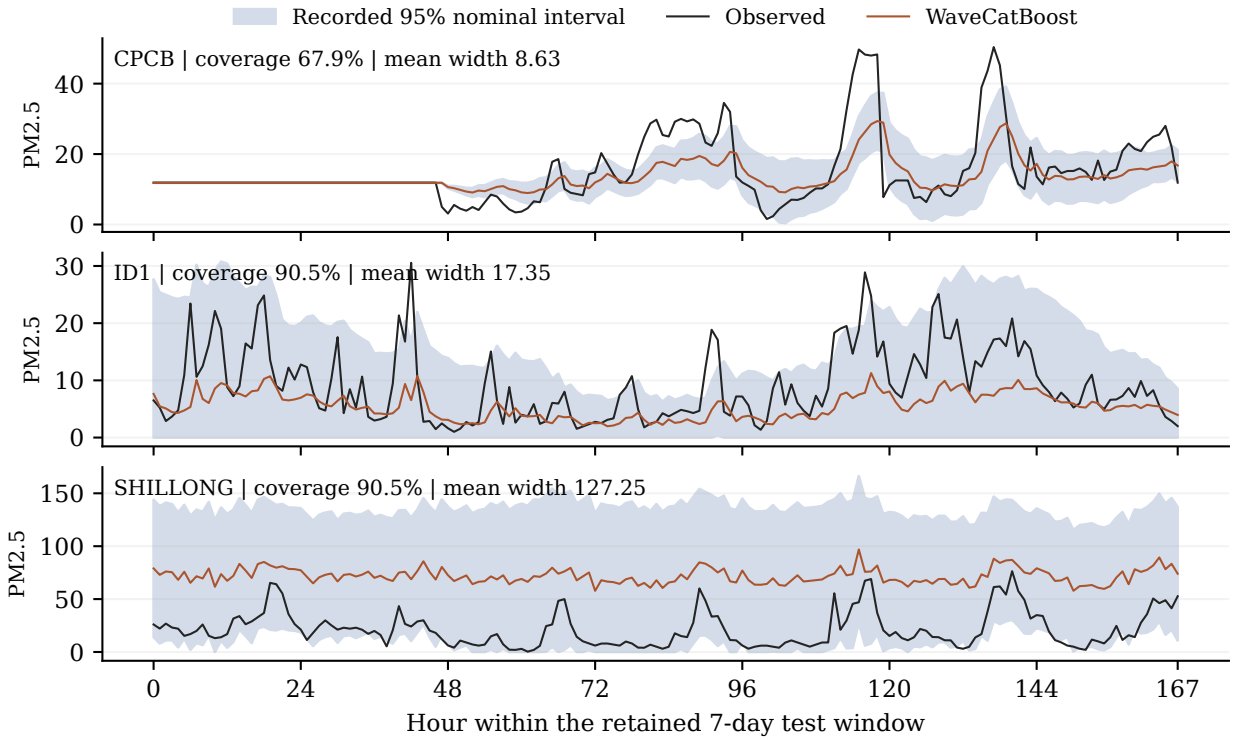


Figure 4. Current saved rolling one-step WaveCatBoost predictions and interval bounds. Each sensor uses its own vertical scale. The shaded bands are from the separate calibration cache, rather than a quantile fitted to the plotted test residuals. Coverage and widths are empirical summaries of the displayed bounds.

.com/jborah2/WaveCatBoost-Time-Series-Model. Companion repository of Borah et al.

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